

Research paper

Control of DC-DC boost converter in discontinuous conduction mode feeding a constant power load

Federico M. Serra^{a,*}, Francisco D. Esteban^a, Oscar Danilo Montoya^b

^a Laboratorio de Control Automático (LCA), Facultad de Ingeniería y Ciencias Agropecuarias, Universidad Nacional de San Luis, CONICET, Villa Mercedes, San Luis, Argentina

^b Grupo de Compatibilidad e Interferencia Electromagnética (GCEM), Facultad de Ingeniería, Universidad Distrital Francisco José de Caldas, Bogotá 110231, Colombia

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ABSTRACT

This article presents a new control strategy for a DC-DC boost converter used to supply a constant power load. The designed controller allows regulates the converter output voltage to a required reference value. By designing the controller based on the reduced order averaged model of the converter and using a nonlinear control technique based on feedback linearization, a simple-to-implement and stable controller for both operating modes of the converter (continuous and discontinuous conduction modes respectively) is obtained. The designed control strategy is validated through simulation and experimental results, demonstrating good dynamic performance and stability in both conduction modes, under variations in reference values and significant load changes. In addition, a comparison of the performance and implementation costs between the control strategy proposed in this paper and two strategies proposed by other authors is included.

1. Introduction

Currently, power electronic converters are gaining increasing relevance as they help meet the growing energy demand and contribute to the reduction of greenhouse gas emissions since they play a fundamental role in integrating renewable energy sources into the electrical grid and in the development of electric vehicles [1–4].

This is mainly because power converters enable the interconnection of different types of systems, regardless of the type of current and the voltage levels at which each operates [5,6]. Particularly, a widely used conversion topology is the boost DC-DC converter, as it allows the elevation of DC voltage. It is employed in applications such as battery chargers for traction or energy storage systems [7,8], photovoltaic voltage regulators [9,10], power factor correction [11–13], distributed generation systems [14], and microgrid applications [15,16], among others.

Based on the current in its inductor, this converter can operate in two modes: (1) Continuous Conduction Mode (CCM) and (2) Discontinuous Conduction Mode (DCM). CCM occurs when the current in the inductor does not reach zero during a switching period, while DCM occurs when the current in the inductor reaches zero before the end of the switching period [17].

Due to its capability of good regulation and low ripple in the output voltage, CCM is often employed in numerous applications, such as fuel cells and electric vehicles [18,19], among others. While DCM is preferred in low-power applications, such as biomedicine, nano, or micro electromechanics systems, and power factor correctors [20,21], because it allows reducing losses in the converter and also reducing the size of the inductor, thereby lowering the system cost [22].

Due to the wide variety of applications of the boost converter operating in CCM, numerous studies in the literature have focused on obtaining mathematical models and designing control strategies. Regarding mathematical modeling, in [23] a classical averaged model is proposed that considers conduction and switching losses in semiconductor devices, resulting in a more accurate model. Meanwhile, in [24], a generalized averaged model for multi-phase DC-DC buck, boost, and buck-boost converters is developed, thereby achieving a mathematical model that describes the DC and AC behavior of the output voltage and inductor current, although it is more complex compared to the classical averaged model.

In [25] a linear control strategy based on a proportional integral derivative (PID) controller with Firefly algorithm is presented for a boost converter that powers a brushless direct current (BLDC) motor through a photovoltaic panel. The controller manages to regulate the

* Corresponding author.

E-mail address: fmserra@unsl.edu.ar (F.M. Serra).

Nomenclature

DC	Direct Current
CCM	Continuous Conduction Mode
CPL	Constant Power Load
v_{dc}	Input voltage
i	Inductor current
T	Period of the switching signal
PWM	Pulse Width Modulation
i_D	Current diode
k_p	Proportional gain
L	Inductance of the inductor
P	Power of the CPL

AC	Altern Current
DCM	Discontinuous Conduction Mode
ROAM	Reduced Order Average Model
v_0	Output voltage
u	Control signal
f_s	Frequency of the switching signal
α	Index modulation of the control signal
β	Auxiliary control input
k_i	Integral gain
C	Capacitance of the output capacitor
R_L	Internal resistance of the inductor

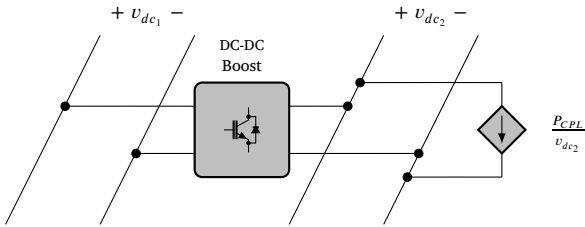


Fig. 1. DC-DC boost converter supplying a CPL.

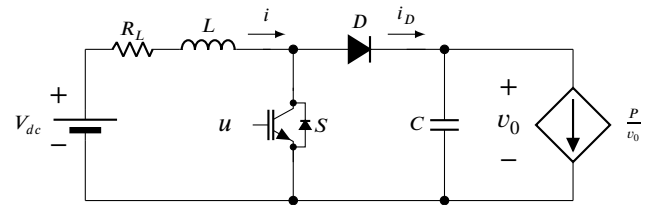


Fig. 2. Scheme of the DC-DC boost converter with CPL.

output voltage of the converter, reducing oscillations and settling time. On the other hand, in [26], a linear H_∞ based controller is proposed to regulate the output voltage of a boost converter supplying a resistive load. This controller is designed to be robust against load variations. Additionally, in [27], a linear controller based on Robust Model Reference Adaptive Control (MRAC) is proposed. This controller achieves robustness against uncertainties and parametric variations. These control approaches achieve low complexity controllers, although they may guarantee good performance and stability when the converter deviates from its operating point, since their design depends on the operating point considered in the linearization.

To address this issue, nonlinear control strategies have also been developed. In [28], a sliding mode controller is proposed, capable of tracking reference voltage in the presence of uncertainties due to variations in input voltage, load resistance, and even reference voltage. However, it is only evaluated with resistive load. In [29], a feedback linearization (FL) based controller is designed for a boost converter that supplies a battery bank from a photovoltaic system, achieving a stable controller under irradiance changes. Meanwhile, in [30], a passivity-based control strategy using interconnection and damping assignment (IDA-PBC) is designed to regulate the output voltage of a boost converter supplying an unknown constant power load (CPL), achieving a globally stable controller that regulates the output voltage against simultaneous variations in input voltage and CPL power.

Although the aforementioned control strategies can regulate the converter output voltage under variations in input voltage, output voltage, and load, they do not ensure voltage regulation when the load exhibits large variations in demanded power, as the converter may operate in DCM under these conditions. This scenario can occur when the converter supplies a CPL, as shown in Fig. 1, which exhibits large variations in demanded power. Therefore, the development of control strategies to regulate the output voltage in a boost converter capable of operating in both CCM and DCM is a topic of great interest among specialists [31].

To this end, several studies in the literature have proposed control strategies for both operating modes of the converter. In [21], a nonlinear Gain-Scheduling-based controller is designed to regulate the output voltage of a boost converter supplying a CPL. The result is a controller that is simple to implement, although its parameters depend on the operating mode of the converter and its physical parameters. In [32], a

strategy is designed to regulate the output voltage in a boost converter based on IDA-PBC, resulting in a globally stable controller. However, this approach requires measurement of the output voltage, inductor and output currents, knowledge of the inductor current time intervals, and two control loops.

Based on the literature review conducted earlier, this article proposes the design of a new control strategy to regulate the output voltage of a boost converter supplying a CPL and operating in both CCM and DCM. The main contributions of this work are as follows:

- Because the controller is designed based on the reduced order averaging model (ROAM) of the converter and employs a FL based nonlinear design technique, good dynamic performance is achieved under reference changes, input voltage variations, and large excursions of load power.
- The controller is valid for both operating modes of the converter, and simple to implement, as it does not require high computational resources, and does not need knowledge of the inductor current time intervals nor their measurement (only the measurement of output voltage and current is needed).
- Finally, the performance of the proposed controller was evaluated through simulation and experimental results, and a comparison was made with the control strategies proposed in [32] and [33], assessing the computational and implementation costs of each, as well as the achieved performance in both operating modes of the converter.

This paper is organized as follows. In Section 2 the modeling of the converter is carried out; Section 3 presents the proposed control strategy; Section 4 shows the results obtained, and finally, Section 5 gives the conclusions.

2. Mathematical modeling

The scheme of the DC-DC boost converter considered in this work is shown in Fig. 2, and the waveforms of the inductor current i and control signal u , considering DCM, are shown in Fig. 3.

From Fig. 2 and Fig. 3 the switched model of the converter can be obtained. The resistance of both power devices is neglected, the input voltage is constant, and the switching is considered instantaneous.

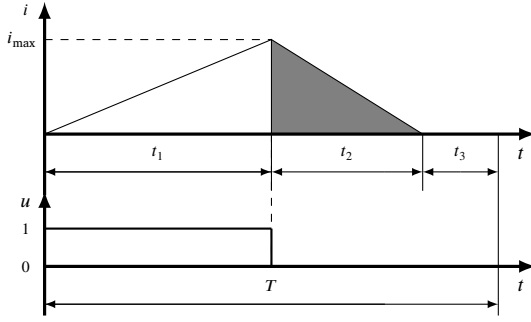


Fig. 3. Waveforms of the inductor current i and control signal u of the DC-DC boost converter in DCM.

$$L\dot{i} = (V_{dc} - R_L i - v_0(1-u)) \left(\frac{1 + \text{sign}(i)}{2} \right), \quad (1)$$

$$C\dot{v}_0 = i(1-u) - \frac{P}{v_0}, \quad (2)$$

being,

$$\text{sign}(i) = \begin{cases} -1 & \text{if } i \leq 0, \\ 1 & \text{if } i > 0, \end{cases} \quad (3)$$

where i represents the inductor current, v_0 and V_{dc} are the output and input voltage respectively, P is the power of the CPL, L is the inductance of the inductor, with internal resistance R_L , and C is the capacitance of the output capacitor. It is important to highlight that if the inductor current remains positive, the model of the DC-DC boost converter for continuous conduction mode is obtained [34].

2.1. Reduced order average model (ROAM)

The ROAM method enables the division of system dynamics into two components: the low-frequency dynamics, which is governed by v_0 , and the fast or high-frequency dynamics, governed by i [34]. The significance of this dynamic separation arises from the involvement of both low and high-frequency signals in the converter. Therefore, a model is needed that remains valid for both conduction modes. By employing the ROAM method, the dynamic behavior of the current can be neglected, treated as a perturbation. This omission allows for a focused approach in controller design, centering attention on voltage dynamics through the consideration of a simplified first-order model.

In order to obtain the ROAM for the DC-DC boost converter considered in this work, R_L is assumed to be zero, given that in a real scenario this value is very small. From Fig. 3 it can be observed that the inductor current is divided into three intervals; in the first one, t_1 , the current i is increased with a slope of V_{dc}/L . In the second interval, t_2 , the current decreases with a slope of $(v_0 - V_{dc})/L$, and finally, in the third interval, the current becomes zero.

Due to the above, and considering $t_1 = \alpha T$, where α is the modulation index of u and T its period, $i_{\max}$ can be obtained as,

$$i_{\max} = \frac{V_{dc}}{L} t_1 = \frac{v_0 - V_{dc}}{L} t_2, \quad (4)$$

t_2 can be calculated from (4),

$$t_2 = \frac{V_{dc}}{v_0 - V_{dc}} \alpha T. \quad (5)$$

The sliding average of i_D related to the commutation period T is,

$$\langle i_D \rangle_0 = \frac{T \alpha^2 V_{dc}^2}{2L(v_0 - V_{dc})}. \quad (6)$$

By averaging the expression (2),

$$C\langle \dot{v}_0 \rangle_0 = \langle i \rangle_0(1 - \alpha) - \frac{P}{\langle v_0 \rangle_0}. \quad (7)$$

Since $\langle i_D \rangle_0 = \langle i \rangle_0(1 - \alpha)$, and replacing (6) in (7), the average value of the output voltage dynamic of the converter is,

$$C\langle \dot{v}_0 \rangle_0 = \frac{T \alpha^2 V_{dc}^2}{2L(\langle v_0 \rangle_0 - V_{dc})} - \frac{P}{\langle v_0 \rangle_0}. \quad (8)$$

Evaluating (8) in steady state, the modulation index for the selected operation point of the converter is determined by,

$$\alpha^e = \sqrt{\frac{2LP(\langle v_0 \rangle_0^d - V_{dc})}{TV_{dc}^2 \langle v_0 \rangle_0^d}}, \quad (9)$$

being $\langle v_0 \rangle_0^d$ the desired value of the output voltage.

3. Control strategy

The proposed controller is designed using feedback linearization control theory. This design technique allows design a linear controller to control a nonlinear system through coordinates transformation and nonlinear feedback [34]. Therefore, this control strategy combines the advantages of nonlinear control techniques (dynamic performance and stability throughout the system's operating range) with the design simplicity offered by linear control techniques.

The output selected for the controller design to regulate the output voltage,

$$z = C\langle v_0 \rangle_0. \quad (10)$$

By taking the derivative of (10) with respect to time, (7) is obtained. As the control input explicitly appears in the first derivative of the output, it becomes possible to define a new linear system with transformed coordinates.

$$\dot{z} = \beta, \quad (11)$$

where β is an auxiliary control input designed to regulate the output voltage in a desired value, given by,

$$\beta = k_p(\langle v_0 \rangle_0^d - \langle v_0 \rangle_0) + k_i \int (\langle v_0 \rangle_0^d - \langle v_0 \rangle_0) dt. \quad (12)$$

Finally, replacing (12) in (7), the control input can be obtained,

$$\alpha = \sqrt{\frac{2L(\langle v_0 \rangle_0 - V_{dc})(\beta + \frac{P}{\langle v_0 \rangle_0})}{V_{dc}^2 T}}. \quad (13)$$

Fig. 4 illustrates the resulting control scheme, comprising a voltage control loop (12) from which the auxiliary control action β is derived. This control action is used in the feedback loop (13) to determine the modulation index α which is then employed in the Pulse Width Modulation (PWM) block. It is important to highlight that, for the implementation of the proposed controller, both the output voltage and current need to be measured (the latter is necessary to determine the power P consumed by the CPL).

4. Results

4.1. Simulation results

Simulation results were performed using Simulink of Matlab with the parameters outlined in Table 1. The implemented control loop is depicted in Fig. 4) and the gains adjusted to achieve a settling time of 2 ms. Specifically, the values of the gains are $k_p = 0.06$ and $k_i = 50$.

Two simulation tests were performed to verify the performance of the controller proposed in this work and to compare it with the control strategy presented in [33]. In both tests, the following sequence of events was implemented: a change in input voltage from 23 V to 24 V occurred at $t = 10$ ms, followed by a change in the reference output voltage from 36 V to 38 V at $t = 20$ ms. At $t = 30$ ms, there was a change in

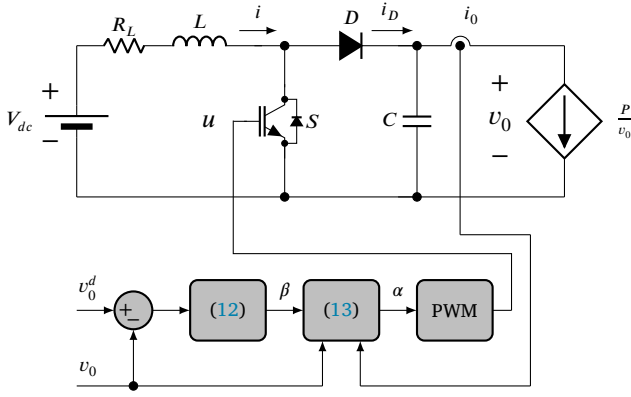


Fig. 4. Proposed control scheme.

Table 1

Converter parameters.

V_{dc}	v_0^d	f_s	L	R_L	C
24 V	36 V	20 kHz	150 μ H	0.2 Ω	150 μ F

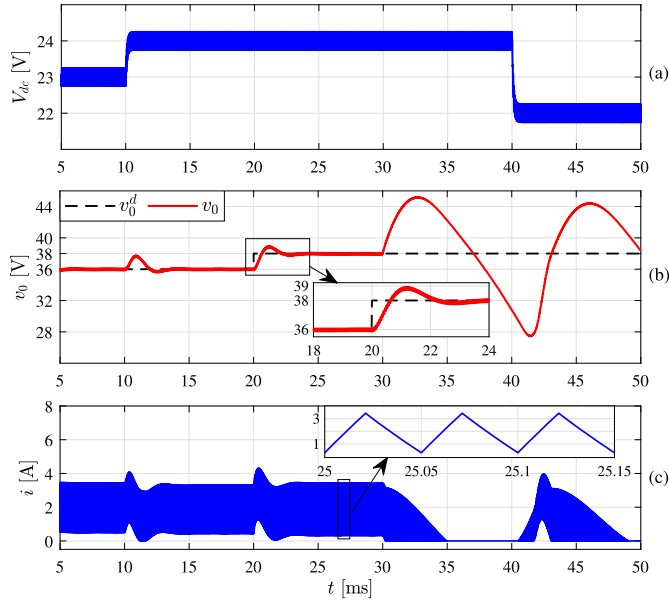


Fig. 5. Performance of the converter when using the controller proposed in [33]. (a) Input voltage. (b) Output voltage. (c) Inductor current.

power of the CPL from 40 W to 10 W, marking the moment when the converter started to operate in DCM. Finally, there was a new change in the input voltage from 24 V to 22 V at $t = 40$ ms.

Fig. 5 illustrates the performance of the controller proposed in [33], whose parameters were adjusted to achieve a response of the output voltage with a settling time of 2 ms. It is observed that, with changes in the input voltage and modifications made to the reference output voltage, the controller maintains the voltage close to its reference value. In both scenarios, the controller exhibits a transient response with overshoots of less than 2 V, minimal oscillation, and a settling time of approximately 2 ms. However, when the power of the CPL decreases to 10 W and the converter starts to operate in DCM, the controller fails to stabilize the output voltage at its reference value.

Fig. 6 shows the performance achieved with the controller proposed in this work. Upon the first change in input voltage, the output voltage exhibits a transient response with an overshoot of 1 V and minimal oscillation, returning to the reference voltage after 2 ms. During the ref-

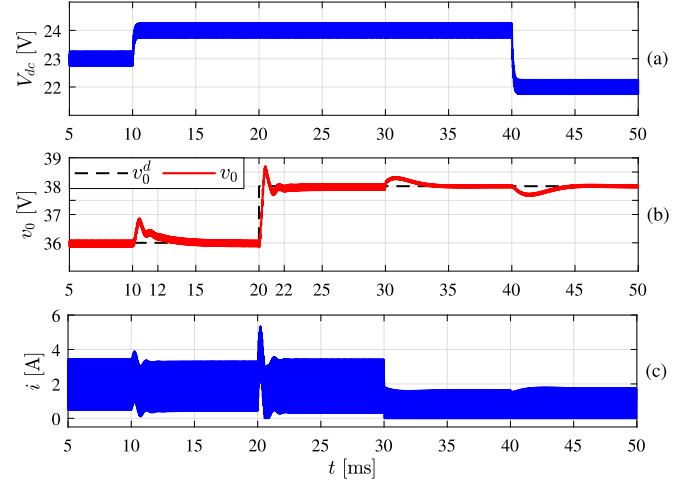


Fig. 6. Performance of the converter when using the controller proposed in this work. (a) Input voltage. (b) Output voltage. (c) Inductor current.

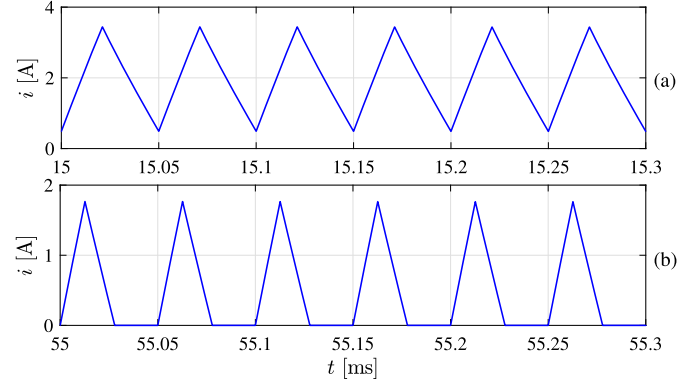


Fig. 7. Current in the inductor in steady state for both modes of operation. (a) In CCM. (b) In DCM.

erence change, an overshoot of approximately 0.75 V is observed, and within 2 ms, the voltage reaches its reference value. Subsequently, when the power of the CPL is reduced to 10 W, the converter begins to operate in DCM. The output voltage experiences an overshoot of 0.5 V and, after the transient, stabilizes at its reference value, similar to the response observed during the second change in input voltage.

Finally, Fig. 7 shows six periods of the steady-state inductor current for both operation modes. Fig. 7 (a) shows the inductor current when the converter operates in CCM, while Fig. 7 (b) shows the inductor current when the converter is operating in DCM.

4.2. Experimental results

Experimental validation of the proposed controller was performed using the laboratory prototype depicted in Fig. 8. The system parameters are listed in Table 1. The control strategy was implemented using a Digital Signal Controller (DSC) Texas Instruments TMS320F28335, with the same gains employed in the simulation tests.

Fig. 9 (a) shows the output voltage and inductor current when utilizing the controller proposed in [33]. A change in the output voltage reference from 36 V to 38 V was executed with a power of 25 W in the CPL. It is evident that in response to the reference change, the voltage reaches its new value within 2 ms, accompanied by a 2 V overshoot.

Fig. 9 (b) illustrates the output voltage and inductor current when the power changes from 25 W to 10 W. It is observed that when the converter begins to operate in DCM, the controller proposed in [33], fails to stabilize the voltage at its reference value.

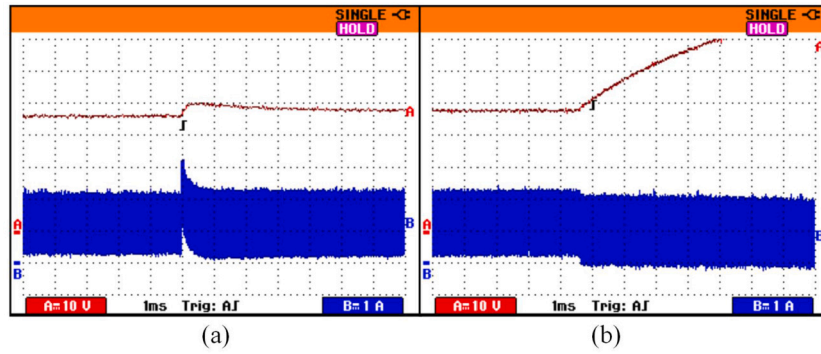


Fig. 9. Output voltage (in red) and inductor current (in blue) when using the controller proposed in [33]. (a) For a change in the voltage reference. (b) For a change in the CPL power.

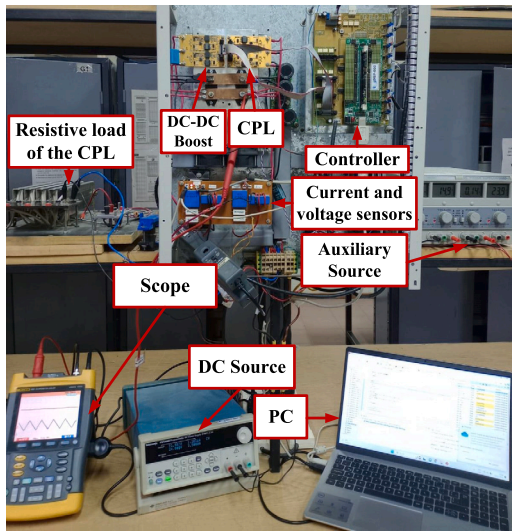


Fig. 8. Image of the laboratory prototype.

In Fig. 10 (a), the output voltage and inductor current are depicted while utilizing the controller proposed in this paper. A voltage reference adjustment is executed, transitioning from 36 V to 38 V with a CPL of 25 W. Notably, the controller effectively maintains voltage regulation at its new set-point, exhibiting a transient overshoot of 2 V and achieving a settling time of 2 ms.

Fig. 10 (b) shows the output voltage and inductor current when the proposed controller in this work is employed, with a change in the CPL from 25 W to 10 W. It is observed that, in response to the change in CPL power, the converter begins to operate in DCM. In this operating mode, the controller successfully regulates the converter output voltage to its reference value, exhibiting a response with a 2 V overshoot.

Finally, Fig. 11 shows the output voltage and inductor current in steady state for both operation modes of the converter. It is observed that the proposed controller effectively maintains the output voltage at the desired set-point across all modes of operation.

4.3. Comparison of control strategies

A comparison is made between the controller proposed in this work and the control strategies proposed in [32] and [33], with the aim of evaluating characteristics such as performance in CCM and DCM, complexity, and computational load of each.

Table 2 shows a comparison between the clock cycles and execution time of the algorithms, with the purpose of evaluating which one uses fewer DSC resources.

Table 2
Comparison of execution times.

Control strategy	Clock cycles	Execution time
Controller proposed in [32]	208	2.77 μ s
Controller proposed in [33]	38	0.5 μ s
Controller proposed in this work	51	0.68 μ s

Finally, Fig. 12 shows a spider chart comparing the three evaluated control strategies. When reading the chart from the inside out, the gray weighting lines indicate poor, fair, and excellent.

The comparison is made according to the following indicators:

- a - Output voltage regulation in CCM: The control strategies proposed in [32] and [33], as well as the controller proposed in this work, achieve excellent output voltage regulation.
- b - Output voltage regulation in DCM: The control strategies proposed in [32] and in this work successfully regulated the output voltage in DCM, whereas the controller proposed in [33] failed to regulate the output voltage.
- c - DSC implementation: Evaluate the number of mathematical operations and the number of sensors required for each control algorithm. The strategy proposed in this work requires the use of two measurement sensors, the calculation of seven algebraic operations, one square root calculation, and the computation of an integral. The controller proposed in [32] requires three measurement sensors, the calculation of thirty-one algebraic operations, and one square root calculation. Lastly, the controller proposed in [33] requires the use of two measurement sensors and the calculation of nine algebraic operations.
- d - Computational burden: Evaluate the time required for the execution of each control algorithm. The computational burden of the controller proposed in [32] is higher compared to the control strategy proposed in [33] and in this work.

5. Conclusion

In this paper, a new control strategy is proposed for a boost converter supplying a CPL. The designed controller effectively regulates the output voltage in both operating modes of the converter (CCM and DCM). Additionally, since the controller is designed based on the ROAM of the converter and employs a nonlinear design technique based on FL, achieving a control law simple to implement and does not require knowledge of the inductor current and their time intervals. The performance of the proposed strategy is validated through simulation and experimental results, demonstrating precise regulation of the output voltage, excellent dynamic performance, robustness, and zero steady-state error for both operating modes of the converter.

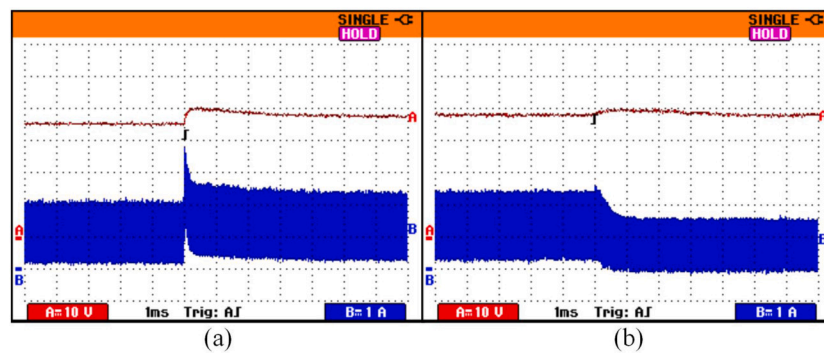


Fig. 10. Output voltage (in red) and inductor current (in blue) when the proposed controller in this work is used. (a) For a change in the reference voltage. (b) For a change in the continuous power load (CPL).

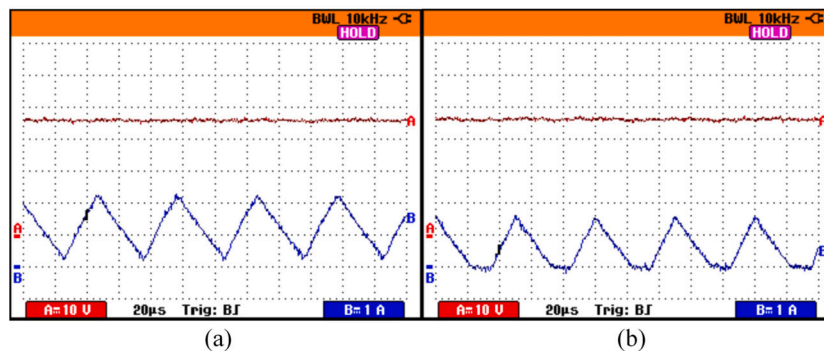


Fig. 11. Output voltage (in red) and inductor current (in blue) in steady state when the proposed controller in this work is used. (a) When the CPL power is 25 W and the converter operates in CCM. (b) When the CPL power is 10 W and the converter operates in DCM.

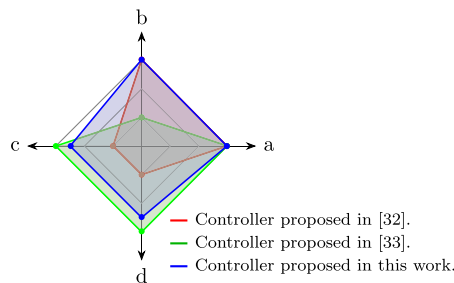


Fig. 12. Spider chart for an illustrative comparison among the controllers proposed in [32], [33] and the controller proposed in this paper.

CRedit authorship contribution statement

Federico M. Serra: Writing – original draft, Visualization, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Francisco D. Esteban:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Conceptualization. **Oscar Danilo Montoya:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] A. Valarmathy, M. Prabhakar, High gain interleaved boost-derived DC-DC converters – a review on structural variations, gain extension mechanisms and applications, *e-Prime, Adv. Electr. Eng. Electron. Energy* 8 (2024) 100618, <https://doi.org/10.1016/j.prime.2024.100618>.
- [2] B.N. Reddy, B.S. Goud, C.N.S. Kalyan, P.K. Balachandran, B. Aljafari, K. Sangeetha, The design of 2S2L-based buck-boost converter with a wide conversion range, *Int. Trans. Electr. Energy Syst.* 2023 (2023) 1–17, <https://doi.org/10.1155/2023/4057091>.
- [3] E. Bushra, K. Zeb, I. Ahmad, M. Khalid, A comprehensive review on recent trends and future prospects of PWM techniques for harmonic suppression in renewable energies based power converters, *Results Eng.* 22 (2024) 102213, <https://doi.org/10.1016/j.rineng.2024.102213>.
- [4] R.M.A. Velásquez, Thermal influence in the design of DC to AC converters due to climatic change for photovoltaic solar plants, *Results Eng.* 23 (2024) 102480, <https://doi.org/10.1016/j.rineng.2024.102480>.
- [5] Z. Alam, S.K. Ghosh, A.F. Alkhateeb, T.K. Roy, M.S. Islam, S. Saha, M.A. Hussain, Robust hybrid nonlinear control approach for stability enhancement of a constant power load DC-DC boost converter, *Alex. Eng. J.* 74 (2023) 535–545, <https://doi.org/10.1016/j.aej.2023.05.041>.
- [6] K. Srilakshmi, N. Yashaswini, S. Gaddameedhi, R. Vangalapudi, P.K. Balachandran, S.R. Salkuti, M.M. Fayaz Basha, Development of AI-controller for solar /battery fed H-bridge cascaded multilevel converter UPQC under different loading conditions, *Int. J. Renew. Energy Res.* 14 (2024) 403–417.
- [7] J. Wang, B. Wang, L. Zhang, J. Wang, N. Shchurov, B. Malozyomov, Review of bidirectional DC-DC converter topologies for hybrid energy storage system of new energy vehicles, *Green Energy Intell. Transp.* 1 (2022) 100010, <https://doi.org/10.1016/j.geits.2022.100010>.
- [8] L. Pirashanthiyah, H.N. Edirisinghe, W.M.P. De Silva, S.R.A. Bolonne, V. Logeeshan, C. Wanigasekara, Design and analysis of a three-phase interleaved DC-DC boost converter with an energy storage system for a PV system, *Energies* 17 (2024) 250, <https://doi.org/10.3390/en17010250>.
- [9] S. Ramasamy, V. Sivasubramaniyam, G. Gatto, A. Kumar, DC link voltage control based energy management strategy for standalone solar PV fed hybrid system, in: 2023 AEIT International Conference on Electrical and Electronic Technologies for Automotive (AEIT AUTOMOTIVE), Modena, Italy, 2023, pp. 1–6, <https://doi.org/10.23919/AEITAUTOMOTIVE58986.2023.10217257>.
- [10] J.S. Oliver, P.W. David, P.K. Balachandran, L. Mihet-Popa, Analysis of grid-interactive PV-fed BLDC pump using optimized MPPT in DC-DC converters, *Sustainability* 14 (2022) 7205, <https://doi.org/10.3390/su14127205>.

- [11] N. Bhati, U. Kalla, Power factor corrected Y-cell modified boost converter fed battery charger for EV applications, *e-Prime, Adv. Electr. Eng. Electron. Energy* (2024) 100550, <https://doi.org/10.1016/j.prime.2024.100550>.
- [12] G. Sarowar, I. Ahmed, S. Rahman, A.A. Mamun, K.M. Salim, Investigation of a power factor correction converter utilizing SEPIC topology with input current switching, *Results Eng.* 22 (2024) 102271, <https://doi.org/10.1016/j.rineng.2024.102271>.
- [13] A.K. Pallekonda, R.K. Ch, High gain interleaved PFC converter for torque ripple minimization in industrial PMLD motor based drives, *Results Eng.* 23 (2024) 102413, <https://doi.org/10.1016/j.rineng.2024.102413>.
- [14] A.S. Al-Khayyat, M.J. Hameed, A.A. Ridha, Optimized power flow control for PV with hybrid energy storage system HESS in low voltage DC microgrid, *e-Prime, Adv. Electr. Eng. Electron. Energy* 6 (2023) 100388, <https://doi.org/10.1016/j.prime.2023.100388>.
- [15] D.A. Palash, Z. Alam, T.K. Roy, A.M.T. Oo, Designing robust hybrid controllers for enhancing output voltage regulation in CPL feed boost converters through backstepping and nonsingular fast terminal sliding mode approaches, *e-Prime, Adv. Electr. Eng. Electron. Energy* 8 (2024) 100532, <https://doi.org/10.1016/j.prime.2024.100532>.
- [16] M.E.T.S. Junior, L.C.G. Freitas, Power electronics for modern sustainable power systems: distributed generation, microgrids and smart grids—a review, *Sustainability* 14 (2022) 3597, <https://doi.org/10.3390/su14063597>.
- [17] F.A. Fairouz, M.A. Elgendy, M. Muhammed, Comprehensive averaged model of DCM-operated boost converter fed by PV source, in: 2024 4th International Conference on Smart Grid and Renewable Energy (SGRE), Doha, Qatar, 2024, pp. 1–6, <https://doi.org/10.1109/SGRE59715.2024.10428988>.
- [18] S. Srividya, P. A, T. Ravi, S. K, Single switch DC-DC boost converter with MPPT control for fuel cell electric vehicle applications, in: 2023 Innovations in Power and Advanced Computing Technologies (i-PACT), Kuala Lumpur, Malaysia, 2023, pp. 1–6, <https://doi.org/10.1109/i-PACT58649.2023.10434573>.
- [19] A. Rajabi, F.M. Shahir, E. Babaei, Performance of a novel DC-DC low voltage stress boost converter for fuel-cell vehicle, *Comput. Electr. Eng.* 111 (2023) 108950, <https://doi.org/10.1016/j.compeleceng.2023.108950>.
- [20] K. Yao, J. Li, F. Shao, B. Zhang, Parallel fixed switching frequency CRM and DCM boost PFC converter with high power factor, *IEEE Trans. Power Electron.* 37 (2022) 3247–3258, <https://doi.org/10.1109/TPEL.2021.3118110>.
- [21] J.G.P. Salado, C.A.H. Ramírez, A.G.S. Sánchez, M.A.R. Licea, Nonlinear stabilization controller for the boost converter with a constant power load in both continuous and discontinuous conduction modes, *Micromachines* 12 (2021) 522, <https://doi.org/10.3390/mi12050522>.
- [22] I. González, A. Sánchez-Squella, D. Langarica-Cordoba, F. Yanine-Misleh, V. Ramirez, A PI + sliding-mode controller based on the discontinuous conduction mode for an unidirectional buck-boost converter with electric vehicle applications, *Energies* 14 (2021) 6785, <https://doi.org/10.3390/en14206785>.
- [23] M. Faifer, L. Piegari, M. Rossi, S. Toscani, An average model of DC-DC step-up converter considering switching losses and parasitic elements, *Energies* 14 (2021) 7780, <https://doi.org/10.3390/en14227780>.
- [24] P. Azer, A. Emadi, Generalized state space average model for multi-phase interleaved buck, boost and buck-boost DC-DC converters: transient, steady-state and switching dynamics, *IEEE Access* 8 (2020) 77735–77745, <https://doi.org/10.1109/ACCESS.2020.2987277>.
- [25] I. Anshory, J. Jamaaluddin, A. Wisaksono, I. Sulistiyowati, Hindarto, B.S. Rintyarna, A. Fudholi, Y.A. Rahman, K. Sopian, Optimization DC-DC boost converter of BLDC motor drive by solar panel using PID and firefly algorithm, *Results Eng.* 21 (2024) 101727, <https://doi.org/10.1016/j.rineng.2023.101727>.
- [26] S.R. Kumar, P. Verma, M.K. Ram, S.R. Somanshu, M.N. Anwar, H_{∞} criterion based PI controller for DC-DC boost converter, in: 2022 IEEE 1st Industrial Electronics Society Annual On-Line Conference (ONCON), Kharagpur, India, 2022, pp. 1–5, <https://doi.org/10.1109/ONCON56984.2022.10126845>.
- [27] C. Yanarates, Z. Zhou, Design and cascade PI controller-based robust model reference adaptive control of DC-DC boost converter, *IEEE Access* 10 (2022) 44909–44922, <https://doi.org/10.1109/ACCESS.2022.3169591>.
- [28] K. Wang, W. Wang, L. Wang, L. Li, An improved SOC control strategy for electric vehicle hybrid energy storage systems, *Energies* 13 (2020) 5297, <https://doi.org/10.3390/en13205297>.
- [29] J.-E. Hernandez-Diez, C.-F. Mendez-Barrios, S.-I. Niculescu, E. Barcenar-Barcenar, A current sensorless delay-based control scheme for MPPT-boost converters in photovoltaic systems, *IEEE Access* 8 (2020) 174449–174462, <https://doi.org/10.1109/ACCESS.2020.3024566>.
- [30] S. Rizzo, W. Gil-González, O.D. Montoya, C. Restrepo, J. Muñoz, Adaptive sensorless PI+passivity-based control of a boost converter supplying an unknown CPL, *Mathematics* 10 (2022) 4321, <https://doi.org/10.3390/math10224321>.
- [31] M.K. AL-Nussairi, R. Bayindir, S. Padmanaban, L. Mihet-Popa, P. Siano, Constant power loads (CPL) with microgrids: problem definition, stability analysis and compensation techniques, *Energies* 10 (2017) 1656, <https://doi.org/10.3390/en10101656>.
- [32] F. Serra, G. Magaldi, L.M. Fernandez, G. Larregay, C.D. Angelo, IDA-PBC controller of a DC-DC boost converter for continuous and discontinuous conduction mode, *IEEE Latin Am. Trans.* 16 (2018) 52–58, <https://doi.org/10.1109/TLA.2018.8291454>.
- [33] F.P. Priya, K. Latha, Feedback linearization control of boost converter, in: 2019 2nd International Conference on Intelligent Computing, Instrumentation and Control Technologies (ICICICT), Kannur, India, 2019, pp. 169–173, <https://doi.org/10.1109/ICICICT46008.2019.8993341>.
- [34] S. Bacha, I. Munteanu, A.I. Bratcu, *Power Electronic Converters Modeling and Control*, Springer, London, 2014.